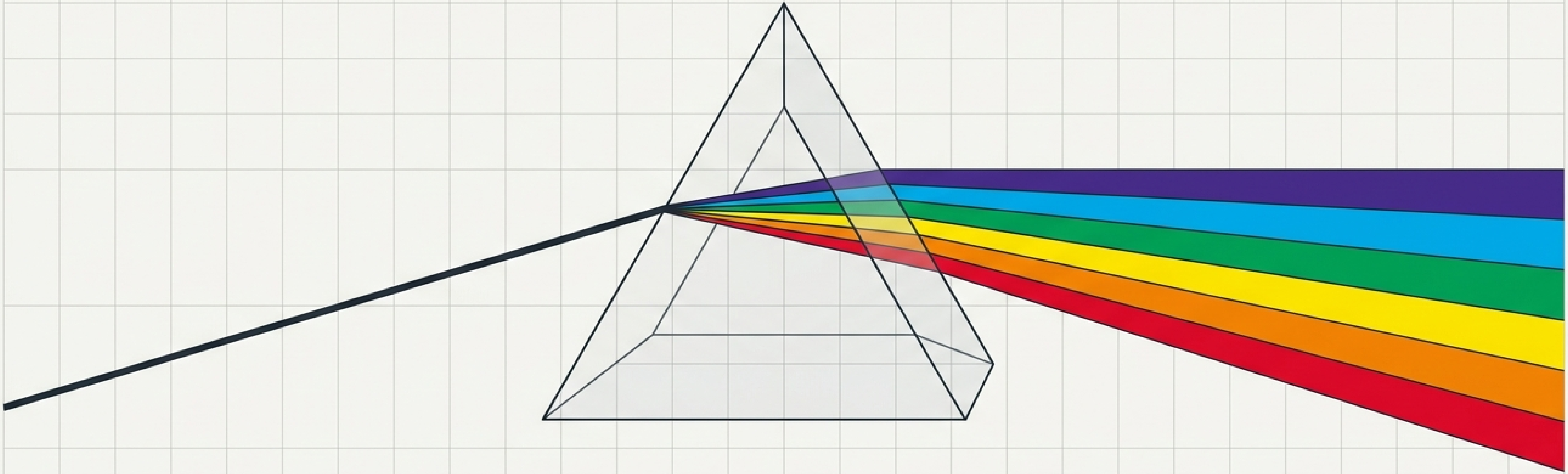
Scale: Universal

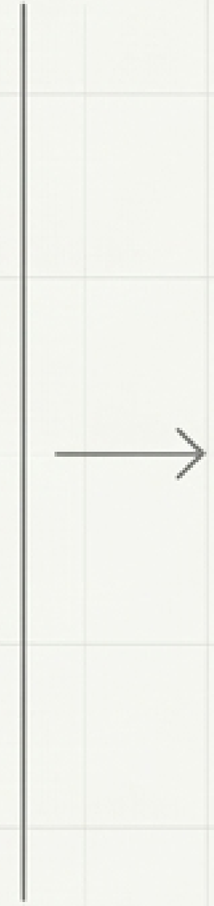


The Wave Nature of Light

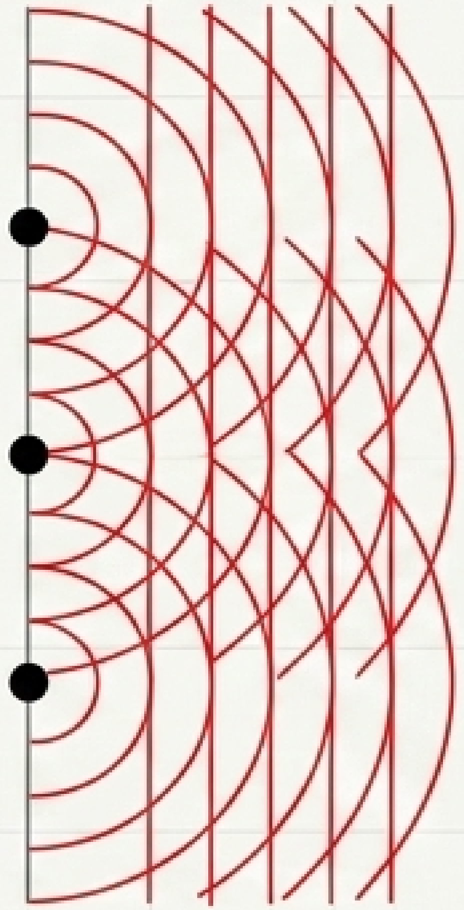
From the colors we perceive to the expansion of the universe.

Light Does Not Travel in Perfectly Straight Lines

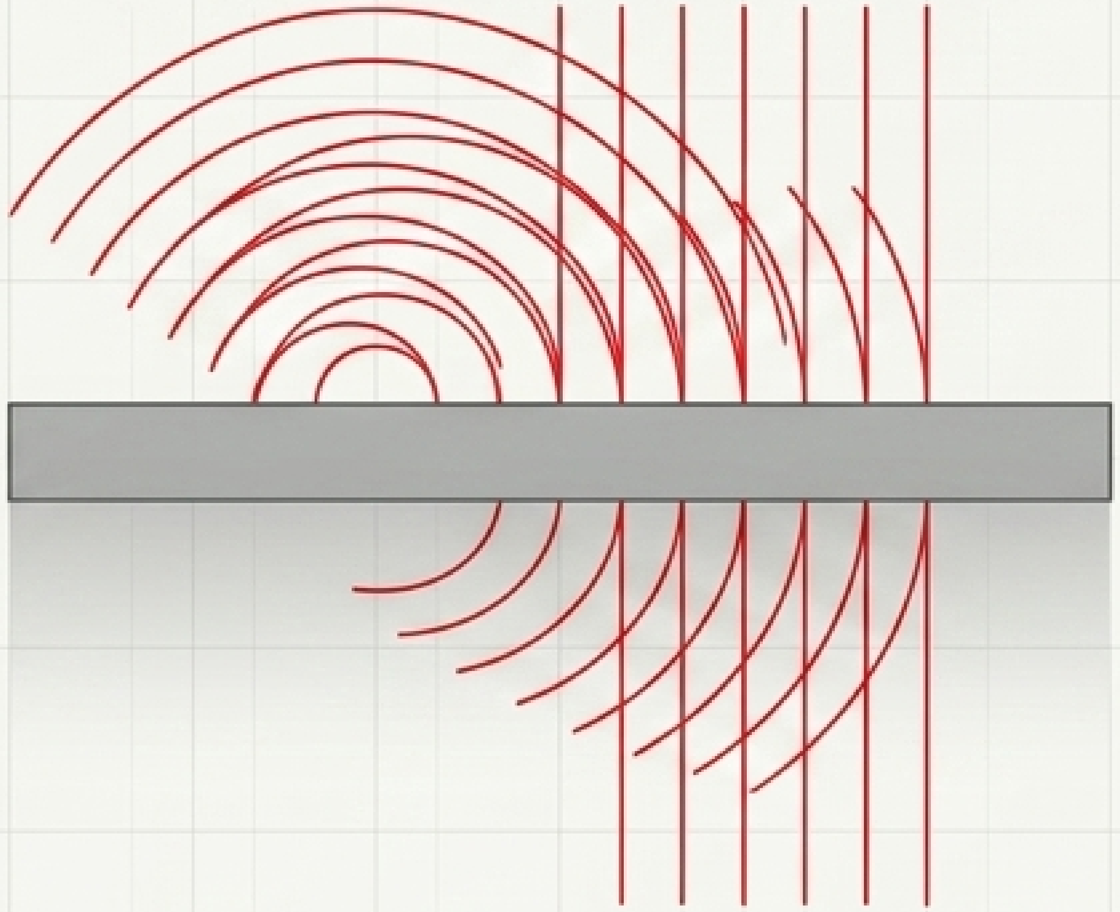
Scale: Fundamental



1. Plane wavefront.



2. Infinite point sources emit expanding wavelets.

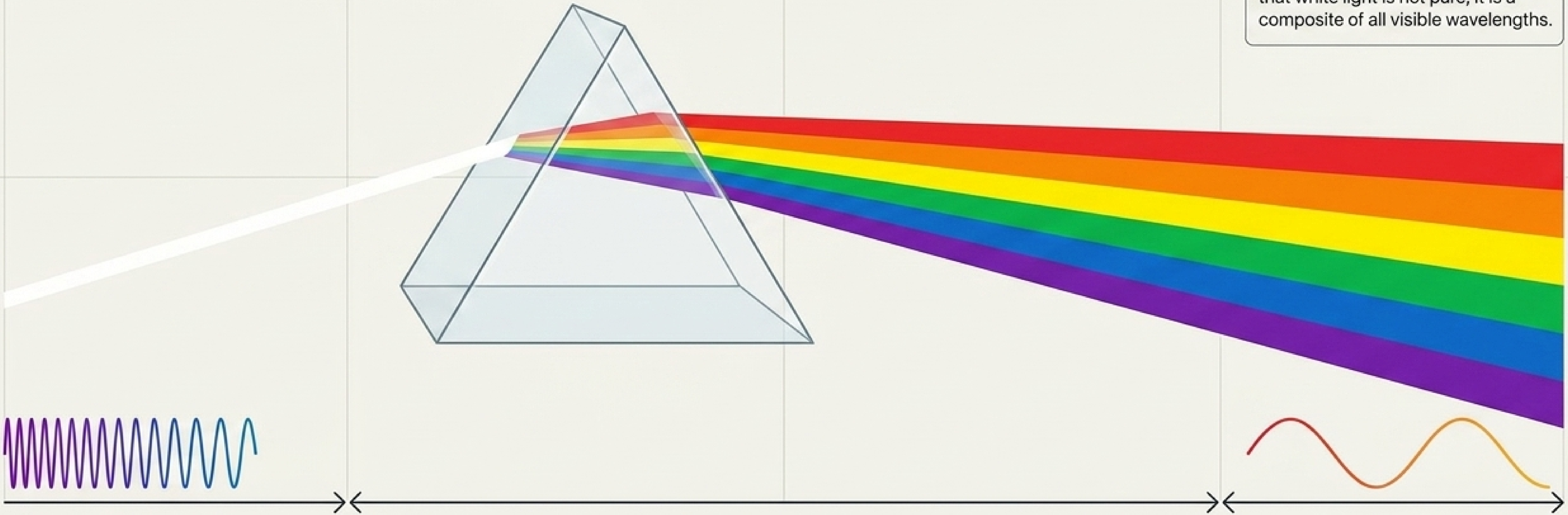


3. Diffraction: Light bends around edges, as observed by Grimaldi (1665).

Key Takeaway: Light is a wave. It propagates by constantly generating new wavelets that combine to push the light forward.

Decoding the Spectrum: Color is Wavelength

Context: Newton (1666) proved that white light is not pure; it is a composite of all visible wavelengths.



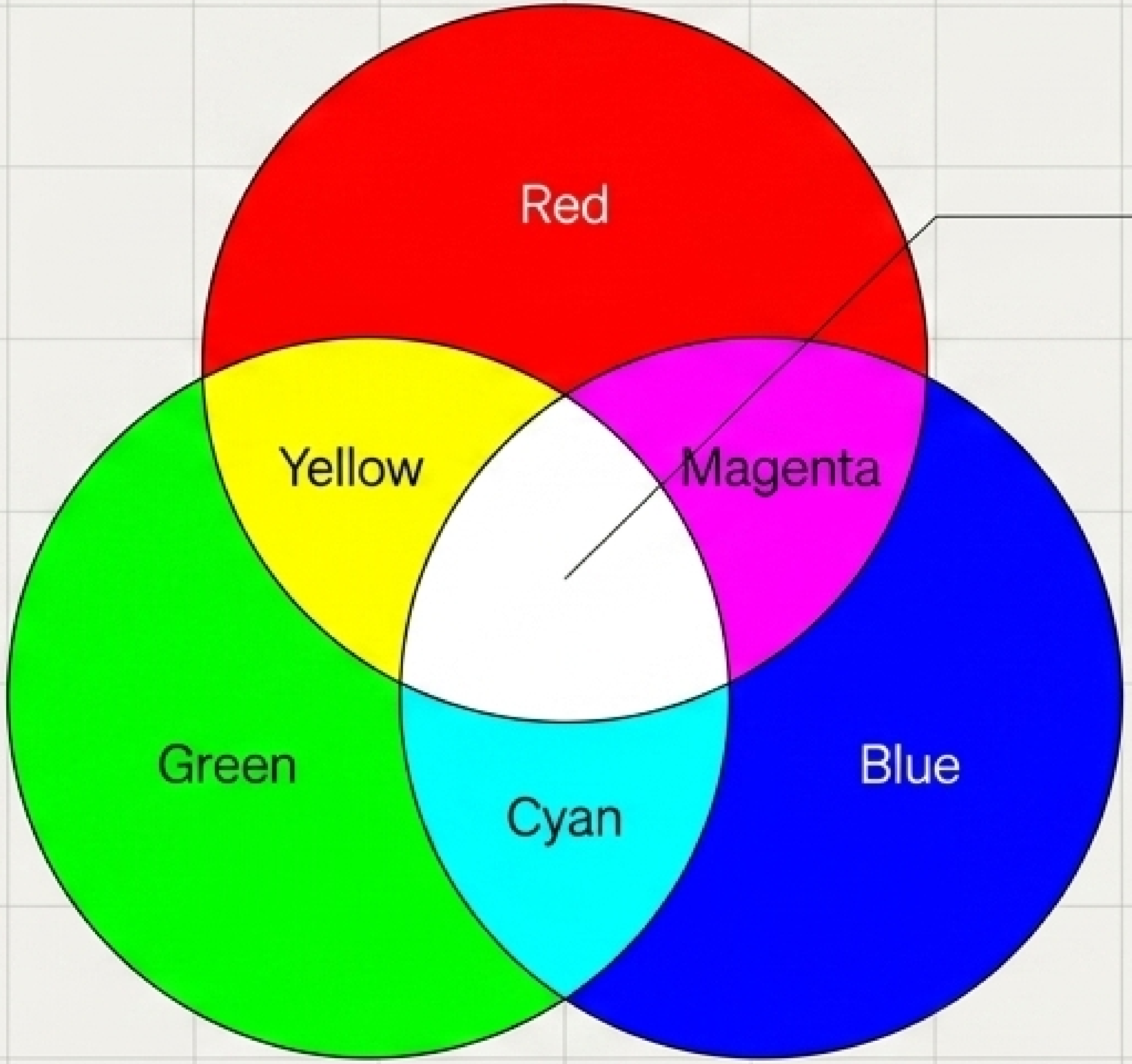
400 nm (4.00×10^{-7} m)
Shorter wavelength = Bends more.

700 nm (7.00×10^{-7} m)
Longer wavelength = Bends less.

The Additive Color Model: Combining Light

Scale: Human Perception

Application:
Used in emitted light technologies, such as television screens and digital projectors.



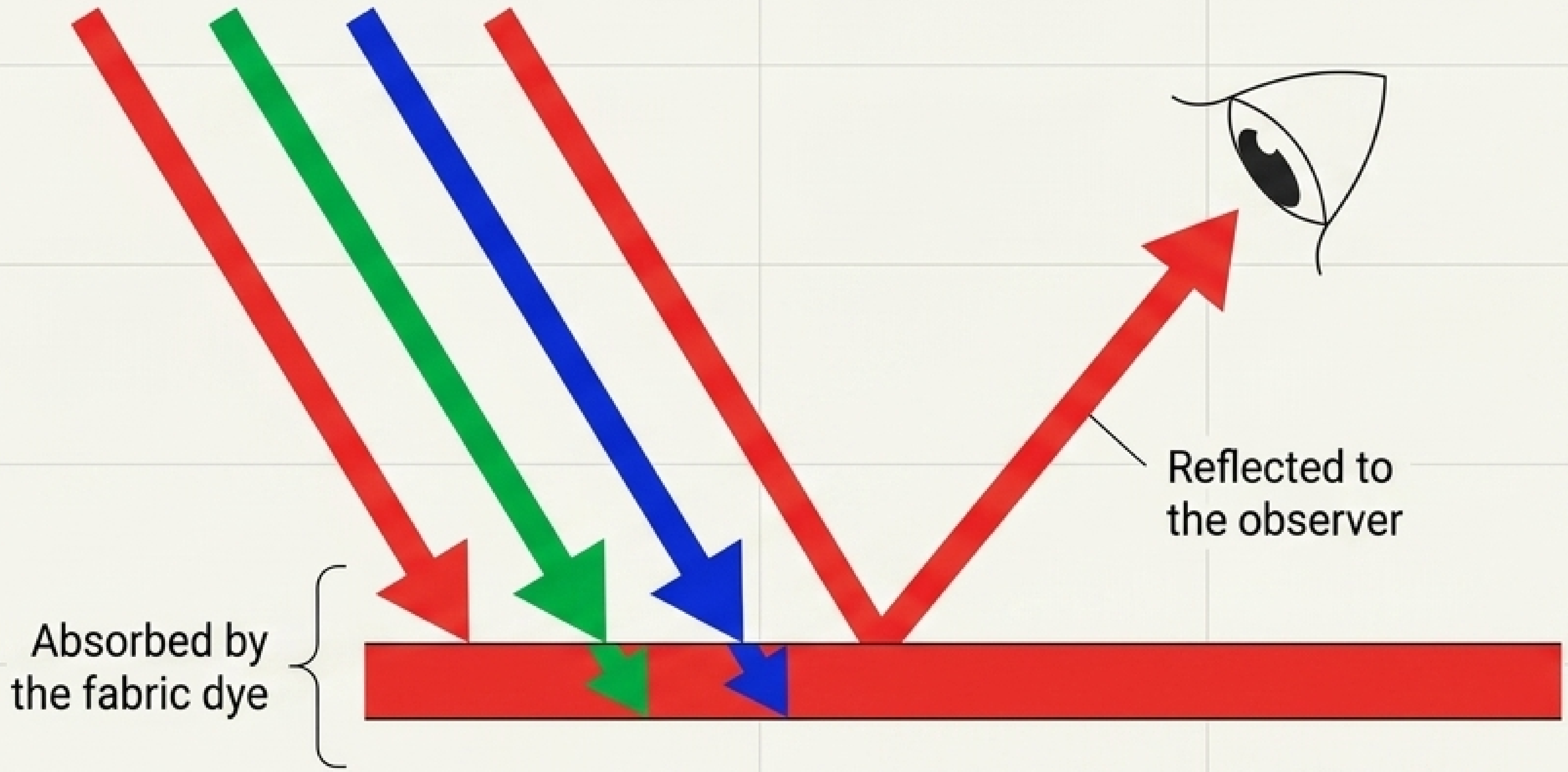
White Light = All visible wavelengths combined.

Complementary Colors
Complementary colors are any two colors of light that combine to make white (e.g., Yellow + Blue).

NotebookLM

The Subtractive Color Model: Absorbing Light

Scale: Human Perception



Sidebar: Pigment Mechanics

- Primary pigments (Cyan, Magenta, Yellow) absorb one primary light color and reflect two.
- Mixing all three primary pigments absorbs all visible wavelengths of light.
- The combination result is black (the absence of reflected light).

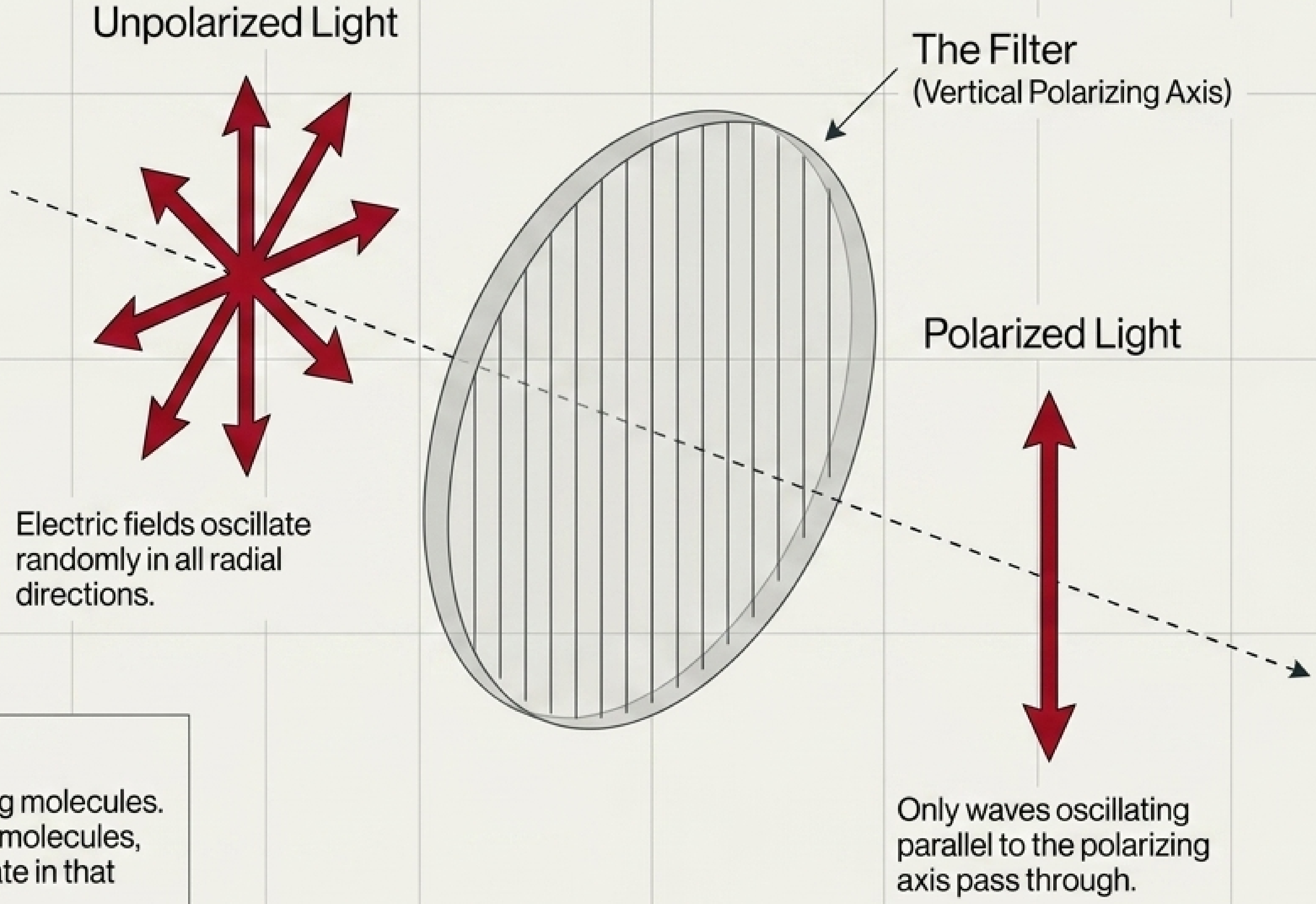
Scale: Human Perception

Two Systems of Color: A Side-by-Side Comparison

	Additive Color (Light)	**Subtractive Color (Pigments/Dyes)**
Medium	Emitted Light (Screens, Projectors)	Physical Matter (Paint, Ink, Leaves)
Mechanism	Adds wavelengths to darkness	Absorbs (subtracts) wavelengths from white light
Primary Colors	Red, Green, Blue (RGB)	Cyan, Magenta, Yellow (CMY)
Secondary Colors	Cyan, Magenta, Yellow	Red, Green, Blue
Combination Result	White Light (All wavelengths present)	Black (All light absorbed)

Polarization: Filtering the Oscillation

Scale: Human Perception

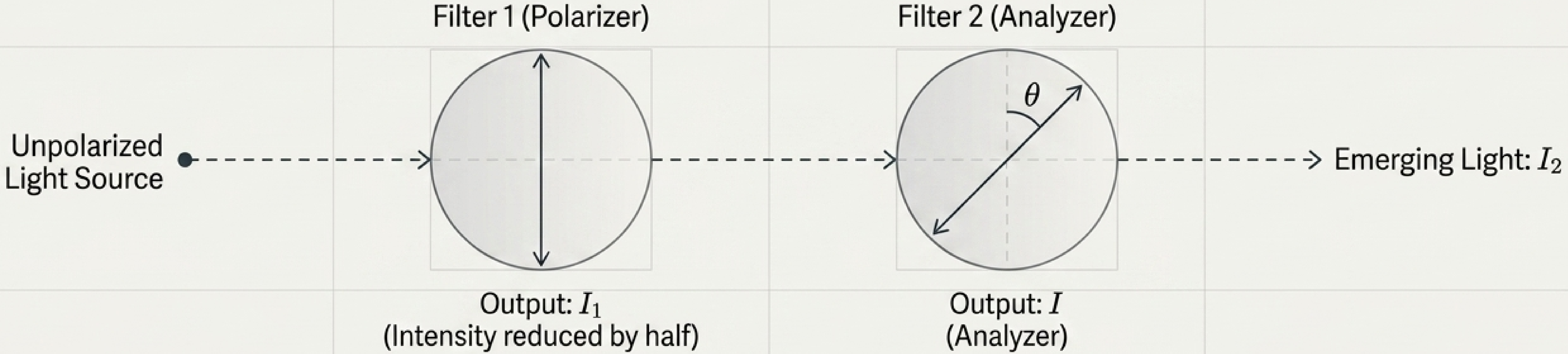


Chemistry Connection

Polarizing mediums contain long molecules. Electrons oscillate along these molecules, absorbing light waves that vibrate in that same direction.

Controlling Intensity: Malus's Law

Scale: Human Perception



$$I_2 = I_1 \cos^2(\theta)$$

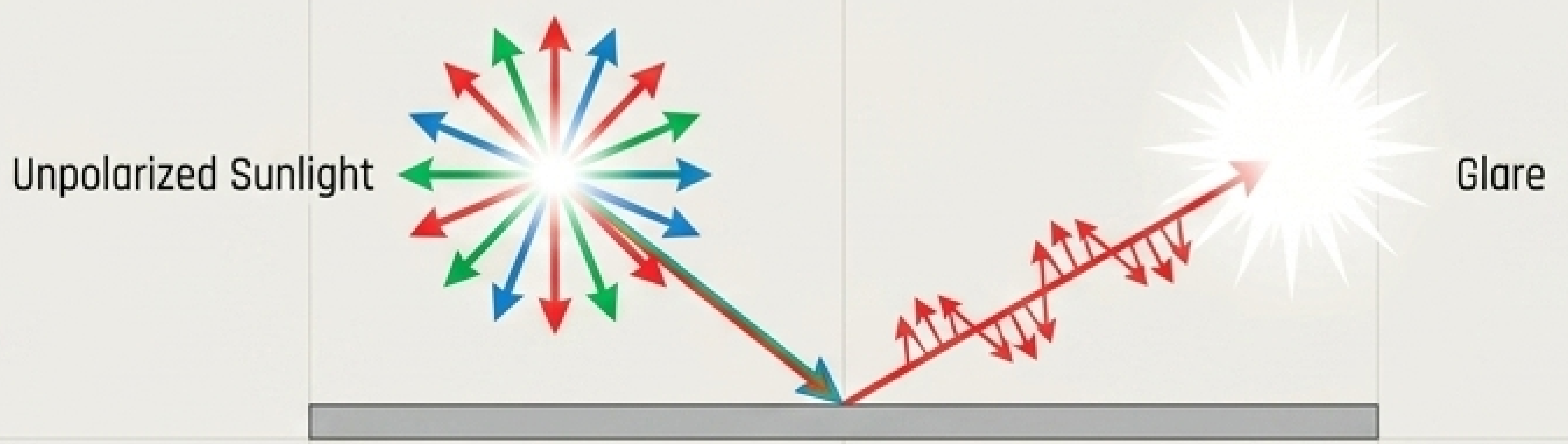
If axes are parallel (θ = 0°), all light passes.
If perpendicular (θ = 90°), zero light passes.

Axes Parallel | Axes at 45° | Axes Perpendicular

Polarization in the Wild: Defeating Glare

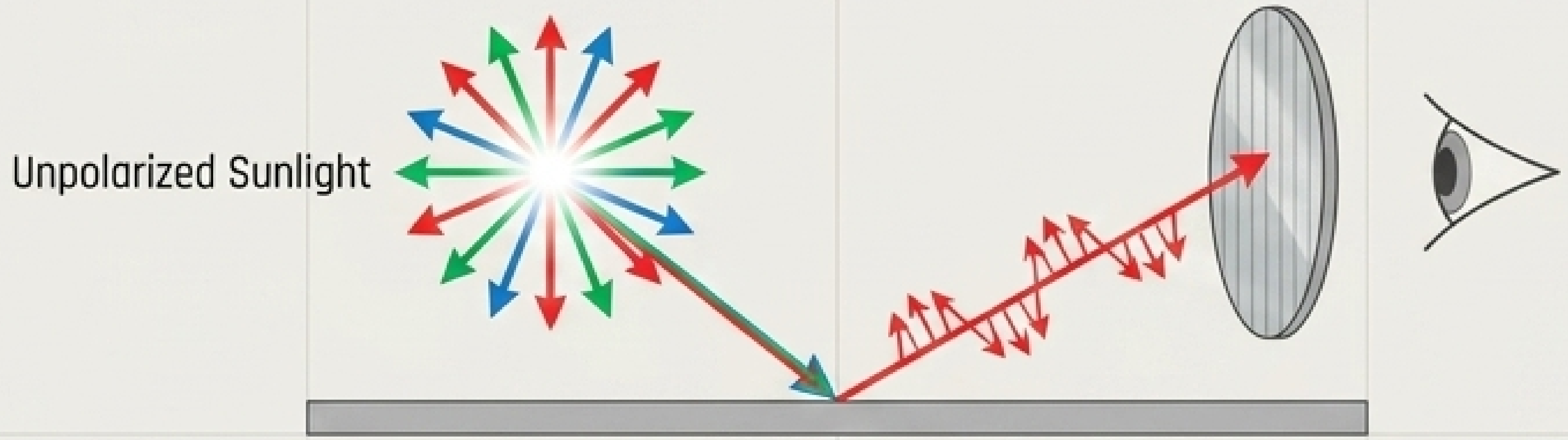
Scale: Human Perception

The Problem



Unpolarized sunlight hits a horizontal surface. Reflected light becomes polarized parallel to the surface, causing blinding glare.

The Solution



Polarizing sunglasses feature a vertical polarizing axis. They completely block the horizontal glare while allowing unpolarized environmental light to pass.

Application: Photographers use polarizing filters over lenses to cut through water and glass reflections.

The Doppler Effect for Light

Scale: Cosmic

Speed of light (constant).

Observed wavelength.

$$\lambda_{\text{obs}} - \lambda = \pm(v/c)\lambda$$

Actual wavelength
generated by the source.

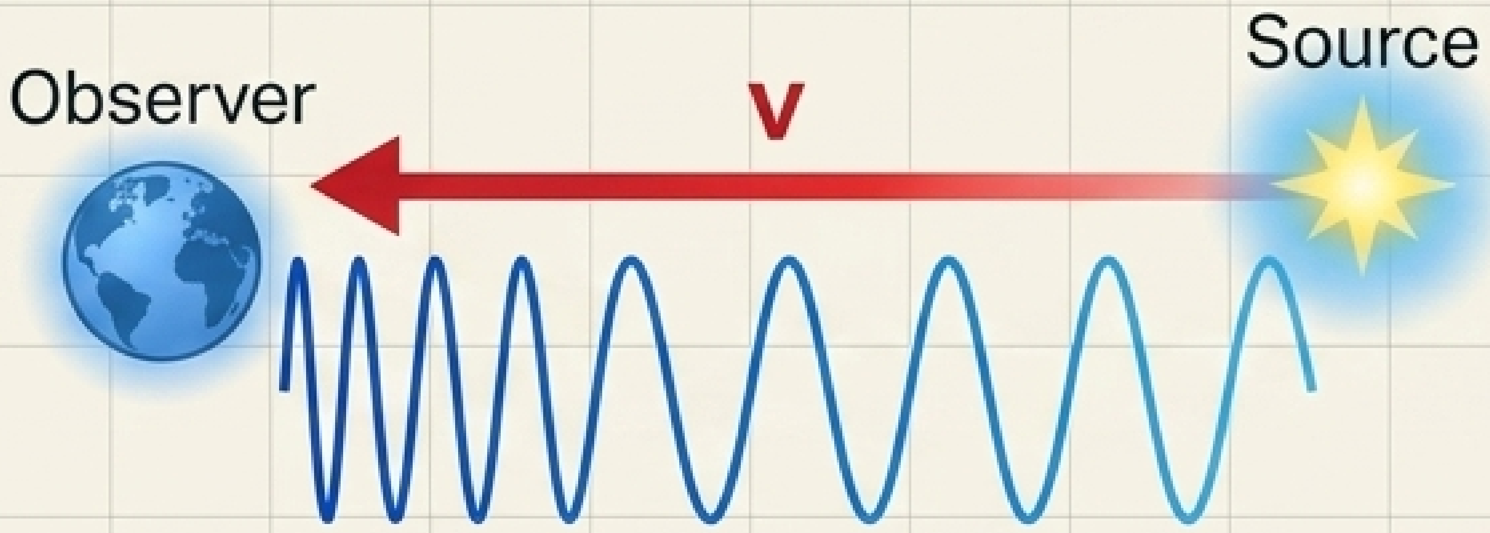
Relative speed between observer
and source along the axis.

Core Concept: The speed of light (*c*) in a vacuum never changes. Since $c = \lambda f$, if the relative motion of a star changes the frequency (*f*) we observe, the wavelength (λ) must stretch or compress in response.

Relative Motion: Stretching and Compressing Waves

Scale: Cosmic

Blueshift (Moving Toward)



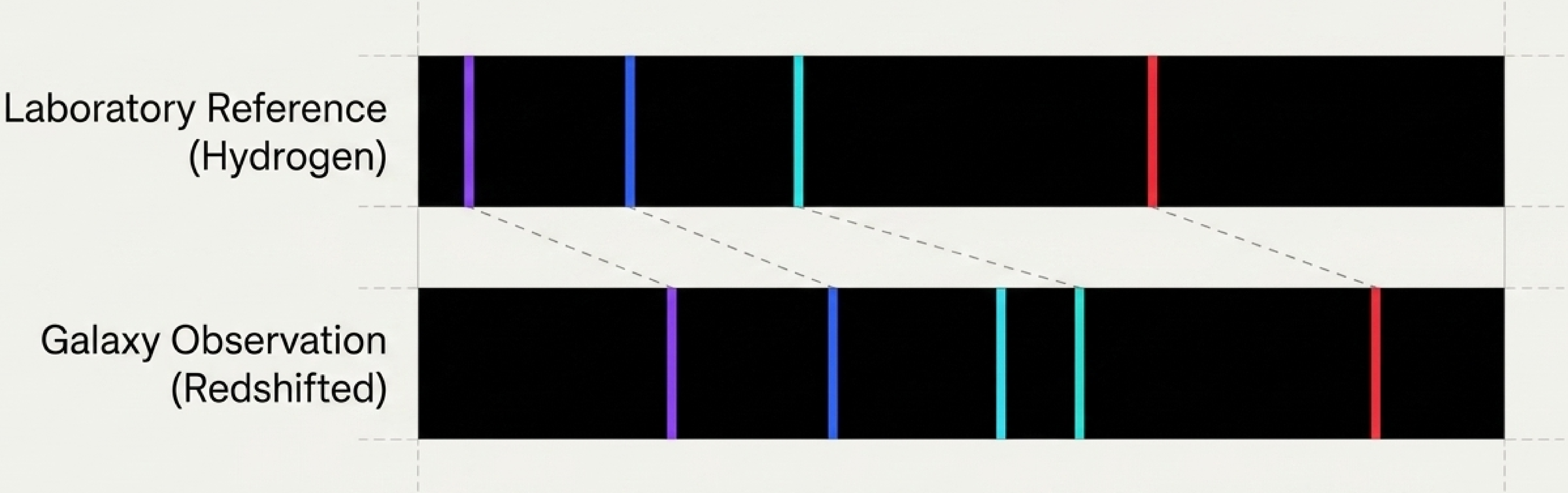
Negative change in wavelength. Waves are compressed. Light shifts toward the shorter, blue end of the spectrum.

Redshift (Moving Away)

Positive change in wavelength. Waves are stretched. Light shifts toward the longer, red end of the spectrum.

Scale: Cosmic

Reading the Spectral Barcode



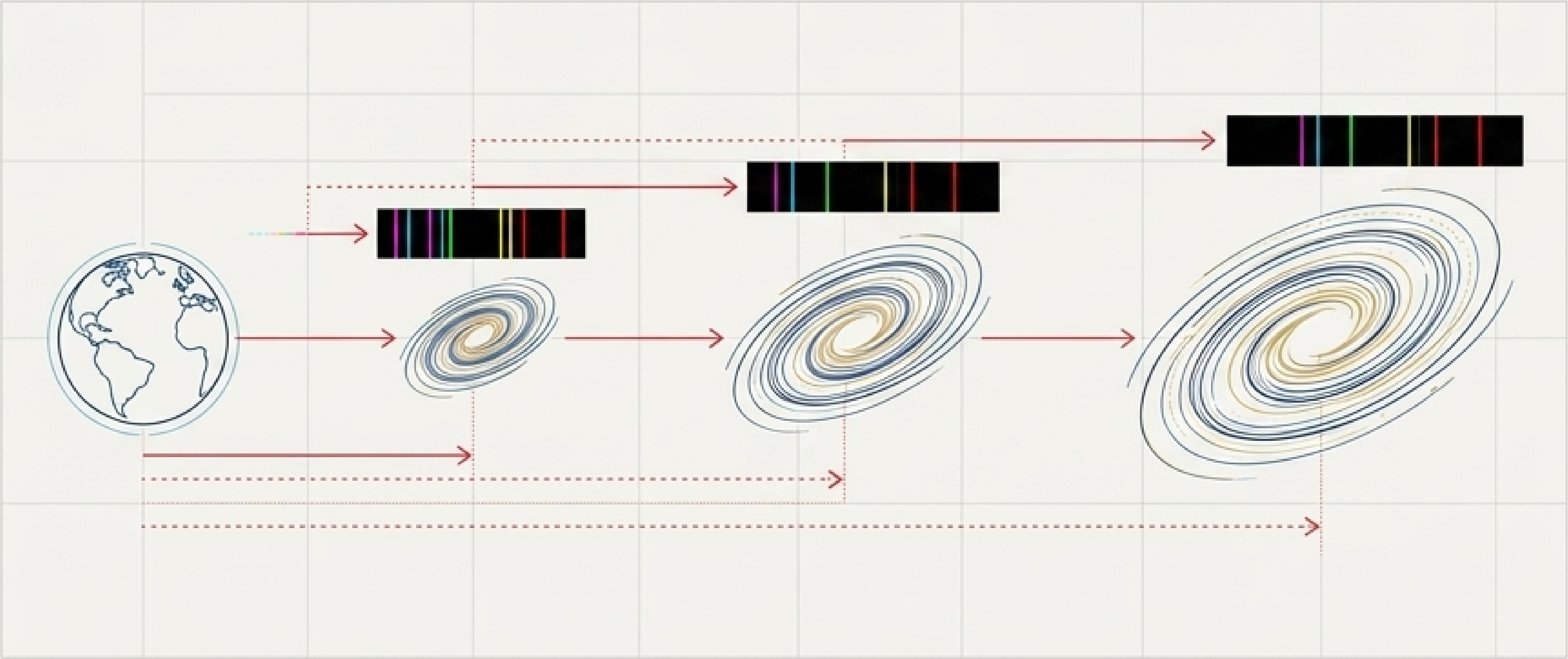
Elements emit light at specific, signature wavelengths. By comparing a static lab spectrum to the light captured from distant galaxies, astronomers can precisely measure the redshift distance and calculate velocity.

Scale: Universal

Hubble's Synthesis: The Expanding Universe

In 1929, Edwin Hubble analyzed light from distant galaxies and found a stunning pattern:

1. Almost all galaxies are redshifted (moving away from Earth).
2. The further away a galaxy is, the greater its redshift.



The Master Key: By understanding **light as a wave**—the same wave mechanics that explain diffraction, color, and glare—we discovered that the universe itself is expanding.